Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

a. Team name: Second Voice
b. Team leader name: Ritik Singh
c. Problem Statement: BFSI. AI to detect and stop fraud that bypasses today's controls: authorised push payment (coercion) scams on UPI, such as digital arrest.

Brief about the idea

Second Voice is a coercion-aware payment interruptor for UPI.

Every bank fraud system asks one question: is this really the account holder? In digital arrest, parcel and investment scams the answer is yes. The victim is on her own phone, typing her own PIN, because a caller posing as a CBI officer told her to. Device, location and credential checks see nothing.

Second Voice sits in the payer bank's authorisation hook, before the debit. It scores whether the account holder is being manipulated right now, from eight signals around the moment of payment, and answers with a Gemini-driven conversation built from the suspected scam script, not a warning banner.

The receiving bank runs the same service on its own accounts and shares one word back during address validation: a receiver risk band. If the customer overrides anyway, the receiving bank holds the credit before it can be cashed out.

THE GAP TODAY

Account takeover

Stolen credentials, new device, odd location. Detected by every bank today.

Authorised push payment fraud

Real customer, real PIN, under coercion. Looks legitimate to every rule. The only intervention in production is a static banner, dismissed in two seconds.

Rising fast in India (digital arrest, task and investment scams). The UK now makes sending and receiving banks reimburse victims 50:50.

Opportunities

HOW IT IS DIFFERENT

Models coercion, not takeover

Existing fraud engines model account takeover. Second Voice models coercion: payee novelty, amount against the customer's own history, an active call, a remote-access app, fan-in from our other customers, and a band from the receiving bank.

The response is a conversation shaped by the suspected scam script, with questions the scammer did not coach the victim for.

HOW IT SOLVES THE PROBLEM

Two banks, two chances

Payer side: pause before the debit, ask, offer a trusted-contact call and a cooling-off, then release, hold or cancel.

Receiver side: the beneficiary bank scores its own accounts (age, KYC depth, first-time inflow fan-in, sweep-out speed) and holds credits to High-band accounts, so money stays inside the banking system even after an override.

USP

A network, not a rule

One field added to the existing UPI address-validation lookup carries the receiver band. Every bank that joins sharpens every other bank's score.

Graduated tiers mean low-risk payments see nothing. Fraud teams get the full reasoning, transcript and a Gemini summary, plus a payee blocklist and I4C report in one click.

List of features offered by the solution

Coercion score

Weighted model over eight interpretable signals, each shown with its contribution. Tiers: low, medium, high, very high.

Script-aware intervention

Gemini names the scam script (digital arrest, parcel, KYC expiry, investment group, task fraud) and runs an adaptive conversation.

Graduated friction

Low: nothing. Medium: one question. High: hold, trusted-contact call, cooling-off. Very high: block and route to a specialist.

Receiver risk band

The receiving bank scores its own accounts and answers address validation with Low, Elevated or High. No account data crosses the wall.

Credit hold at the receiver

An override still lands on a 24-hour hold at the beneficiary bank when the account is High band.

Fraud operations

Live case queue, signals, transcript with the agent's reasoning, Gemini case summary, one-click payee blocklist and I4C report.

Process flow

1 Tap Pay

Bank A app. SDK attaches device context.

2 Address check

Bank A asks Bank B for the name. Bank B answers with name + risk band.

3 Coercion score

History, payee reputation, device, band, registry. Eight signals, one score.

4 Tier

Low: allow. Medium/High: hold and talk. Very high: block.

5 Gemini talks

Script-aware questions. Re-scored on every turn.

OUTCOMES

Release

Customer knows the payee. Payment completes.

Trusted-contact hold

Daughter is called. 24-hour pause. Nothing moves.

Cancel

Customer cancels. Case closed, payee blocklisted if confirmed.

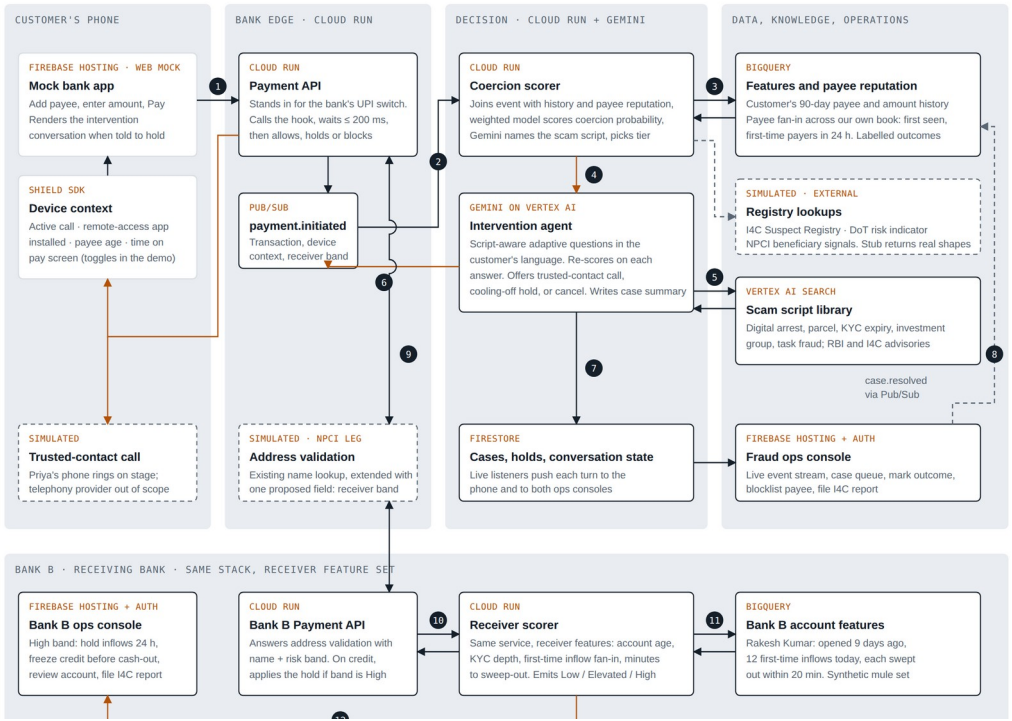
Override

10-second cooling-off, then debit. Bank B holds the credit if High band.

Fraud ops

Transcript, summary, outcome label, I4C report. Outcomes retrain the score.

Architecture



One codebase, two banks

The same Cloud Run service is deployed as bank-a (payer hook) and bank-b (receiver band). Firestore is the only state; live listeners drive the phone and both consoles.

Flow

- 1 App sends pay request + device context
- 2 Payment API validates the address with Bank B, gets the band
- 3 Scorer joins history, fan-in, band, registry; Gemini names the script
- 4 Medium and above: Gemini intervention agent, turns in Firestore
- 5 Decision persists; ops console updates live
- 6 Override: credit to Bank B, held if High band
- 7 Outcomes labelled by ops feed back into the score

Dashed boxes are simulated in the prototype: device signals (toggles), registry lookups, the NPCI leg, the trusted-contact call.

Technologies used in the solution

Cloud Run

Both bank services from one container. Payment API, scorer, Gemini agent, address validation, credit hold. One minimum instance each, asia-south1.

Firestore

Customers, 90-day history, receiver accounts, cases, holds, conversation turns. Composite indexes for the fan-in queries. Live listeners to every screen.

Firebase Hosting + Auth

Mock bank app and two ops consoles as static sites. Anonymous auth gates Firestore reads; all writes come from Cloud Run.

Gemini 2.5 Flash on Vertex AI

Names the scam script from the signal pattern, runs the intervention turns with a per-script system prompt, writes the case summary. JSON-mode responses.

FastAPI, Python 3.12, uv

Service code, weighted scoring model, receiver mule score, Firestore Admin SDK, httpx between banks.

Firebase JS SDK 10

Phone app and consoles in plain HTML/JS; onSnapshot listeners for cases, turns, accounts, holds, events.

Cloud Build, Artifact Registry

Source-to-container deploys through gcloud run deploy --source.

Planned

BigQuery analytics copy of history and outcomes, Pub/Sub event stream, Vertex AI Search over the scam-script corpus, trained model from labelled cases.

Estimated implementation cost

Prototype as deployed (per month, indicative, asia-south1 list prices)

Cloud Run, 2 services, 1 min instance each, 512 MB	~ $14 to $20
Firestore, demo-scale reads and writes	within free tier
Gemini 2.5 Flash, ~3 calls per flagged payment	~ $0.002 per intervention
Firebase Hosting and Auth	free tier
Total	**~ $15 to $25 per month**

At bank scale

Scoring is a few Firestore or BigQuery lookups per payment: compute cost is negligible against existing fraud-engine spend.

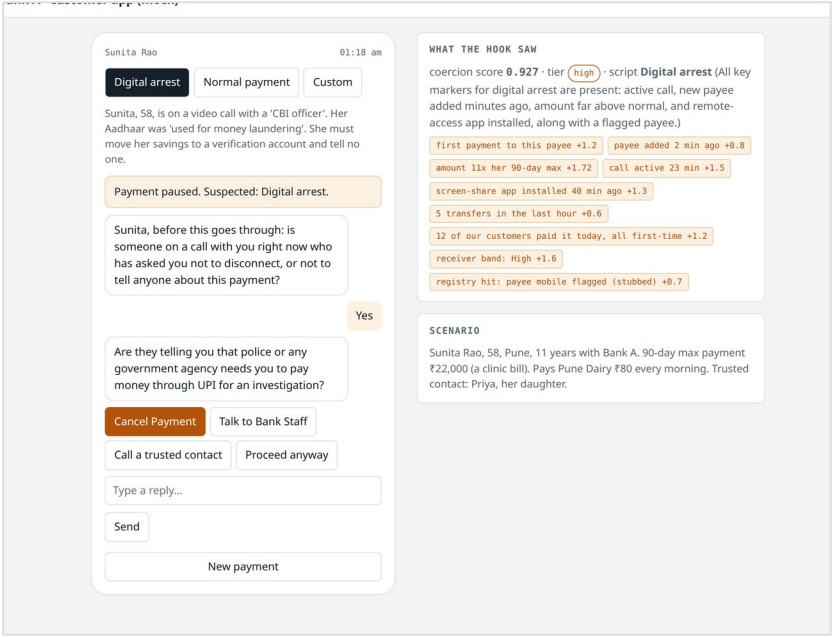
Gemini is called only for the small share of payments that reach medium tier or above. At 1 to 2 percent of volume and three calls per case, a bank doing 10 million UPI payments a month spends on the order of a few hundred dollars on the model.

The real cost is integration: the SDK in the bank's app, the hook in the payment switch, and the band field in address validation.

Set against reimbursement liability for authorised push payment fraud, which the UK now splits between sending and receiving banks.

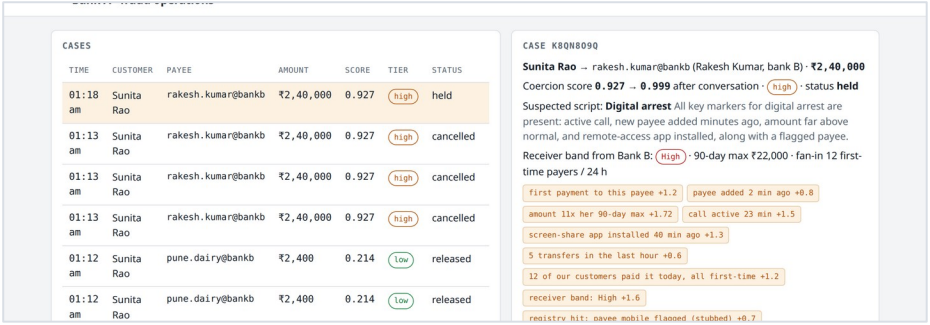
Snapshots of the prototype

PHONE APP: PAYMENT PAUSED, GEMINI CONVERSATION, SIGNALS

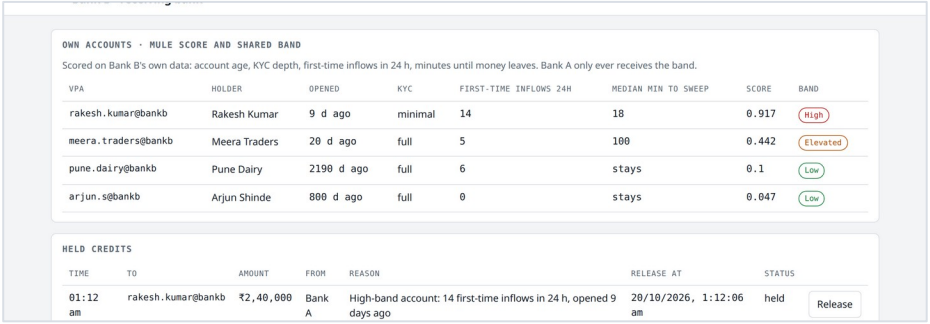


Live at fabled-ray-512016-k5.web.app. No login. Pick Digital arrest, tap Pay.

BANK A FRAUD CONSOLE: CASE, SIGNALS, TRANSCRIPT



BANK B RECEIVER CONSOLE: MULE BANDS, HELD CREDITS



Prototype performance report

Latency on the live deployment, 19 Oct 2026

Service health check	**0.2 s**
Address validation, Bank A to Bank B, with band	**0.23 s**
Low-risk payment: score and allow (no model call)	**0.5 to 0.6 s**
Flagged payment: score, Gemini script classification, hold	**7.5 to 10 s**
Gemini conversation turn	**3 to 5 s**
Receiver credit hold at Bank B	**under 1 s**

Scoring itself is sub-second. The 7 to 10 s on flagged payments is the synchronous Gemini classification; the next revision returns the hold immediately on the rule-based script guess and lets Gemini refine it in the background.

Scoring outcomes on the seeded scenarios

Sunita, digital arrest, ₹2,40,000 to a new payee on a 23-min call	**0.909**	**high**
Sunita, ₹2,400 to her regular milk vendor	**0.178**	**low**
New payee, 3x her usual amount, no call	**0.386**	**medium**
Bank B: Rakesh Kumar, 9 days old, 12 first-time inflows, 18-min sweep	**0.917**	**High**
Bank B: Pune Dairy, 6 years old, full KYC	**0.054**	**Low**
Bank B: Meera Traders, 20 days old, 5 first-time inflows	**0.442**	**Elevated**

Gemini named the script correctly in every live run (digital arrest, confidence 0.95) and produced valid JSON turns and summaries; stub fallbacks never triggered.

Future development

Real device signals

Android SDK reading call state, recently installed remote-access apps and payee age, replacing the demo toggles.

Learned model

Train the coercion score on labelled case outcomes from the ops console; BigQuery as the analytics store, Pub/Sub for the event stream.

Shared registries

Live lookups to the I4C Suspect Registry and the DoT Financial Fraud Risk Indicator; file I4C reports automatically.

Receiver band on the rails

Propose the risk-band field for the UPI address-validation response so any bank pair can exchange it.

Languages and voice

Hindi and Marathi turns; a spoken intervention for customers who cannot read under pressure.

Script corpus

Vertex AI Search over fresh scam scripts and RBI advisories, so the agent learns new patterns without a redeploy.

Links

GitHub public repository

https://github.com/R-I-T-I-K/second-voice

Demo video (3 to 4 minutes)

Demo video link: to be added after upload

Final product link

https://fabled-ray-512016-k5.web.app/

The product link needs no login. Open the Digital arrest scenario and tap Pay; the two consoles update live in other tabs.

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Thank you

